ECON 4720: HA03

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Problems

Problem 1

Suppose that $Y_1, Y_2, \ldots, Y_n$ are random variables with a common mean μ_Y, a common variance σ_Y^2, and the same correlation ρ (so that the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$).

1a

Show that $\text{Cov}(Y_i, Y_j) = \rho\sigma_Y^2$ for $i \neq j$.

Solution Attempt: We know that correlation is defined as

$$Corr(Y_i, Y_j) = \frac{Cov(Y_i, Y_j)}{\sqrt{Var(Y_i)Var(Y_j)}}$$

and from our preamble we also know that "the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$" and common variance implying $Var(Y_i) + Var(Y_j) = \sigma_Y^2$. Thus,

$$\begin{aligned}
\rho &= \frac{Cov(Y_i, Y_j)}{\sqrt{\sigma_Y^2 \sigma_Y^2}} \\
&= \frac{Cov(Y_i, Y_j)}{\sigma_Y^2} \\
\rho\sigma_Y^2 &= Cov(Y_i, Y_j)
\end{aligned}$$

which shows what the question asks.

$\square$

1b

Suppose that $n = 2$. Show that $\text{E}(\bar{Y}) = \mu_Y$ and $\text{Var}(\bar{Y}) = \frac{1}{2}\sigma_Y^2 + \frac{1}{2}\rho\sigma_Y^2$.

Solution Attempt: This change brings our random variable set to $\{Y_1, Y_2\}$. First, we find $E(\bar{Y})$. Note that $\bar{Y}$ is

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i = \frac{1}{2}(Y_1 + Y_2)$$

Thus,

$$E(\bar{Y}) = E\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{2}(E(Y_1) + E(Y_2))$$

From the preamble, we know that $E(Y_i) = E(Y_j) = \mu_Y$, thus,

$$E(\bar{Y}) = \frac{1}{2}(2\mu_Y) = \mu_Y$$

Now, for the variance, $Var(\bar{Y})$. From earlier, we know

$$\begin{aligned}
Var(\bar{Y}) = Var\left(\frac{1}{2}(Y_1 + Y_2)\right) &= \frac{1}{4}Var(Y_1 + Y_2) \\
&= \frac{1}{4}(Var(Y_1) + Var(Y_2) + 2Cov(Y_1, Y_2)) && \text{Expansion} \\
&= \frac{1}{4}(2\sigma_Y^2 + 2\rho\sigma_Y^2) && \text{From preamble and (a)} \\
&= \frac{1}{2}\mu_Y^2 + \frac{1}{2}\rho\sigma_Y^2 && \text{Shows what we want}
\end{aligned}$$

□

1c

For $n \geq 2$, show that $\mathrm{E}(\bar{Y}) = \mu_Y$ and $\mathrm{Var}(\bar{Y}) = \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right)\rho\sigma_Y^2$.

Solution Attempt: Now our set of random variables is {Y_1, Y_2, …, Y_n}. The expectation of $\bar{Y}$ follows a similar process as with (b).

$$\begin{aligned}
\bar{Y} &= \frac{1}{n}\left(Y_1 + Y_2 + \cdots + Y_n\right) = \frac{1}{n} \cdot n\mu_Y = \mu_Y \\
E(\bar{Y}) &= E\left(\frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)\right) = \frac{1}{n}(\underbrace{E(Y_1) + E(Y_2) + \cdots + E(Y_n)}_{n \text{ terms}}) \\
&= \frac{1}{n}(n \cdot \mu_Y) = \mu_Y
\end{aligned}$$

Note that we used the same principle that $E(Y_i) = E(Y_j) = \mu_Y$ to accomplish the above. Now to show the variance of $\bar{Y}$.

$$
\begin{aligned}
Var(\bar{Y}) = Var\left(\frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)\right) &= \frac{1}{n^2}Var(Y_1 + Y_2 + \cdots + Y_n) \\
&= \frac{1}{n^2}\Big(\overbrace{Var(Y_1) + Var(Y_2) + \cdots + Var(Y_n)}^{n \text{ terms}} + \overbrace{2Cov(Y_1, Y_2) + \cdots + 2Cov(Y_1, Y_n)}^{n \text{ choose 2 terms}}\Big) \\
&= \frac{1}{n^2} \cdot n\sigma_Y^2 + \frac{1}{n^2}\binom{n}{2}2\rho\sigma_Y^2 \\
&= \frac{\sigma_Y^2}{n} + \frac{1}{n^2} \cdot \frac{n(n-1)}{2}2\rho\sigma_Y^2 \\
&= \frac{\sigma_Y^2}{n} + \frac{n-1}{n}\rho\sigma_Y^2
\end{aligned}
$$

We have n choose 2 terms because this is every pairwise correlation and that's how many there exist. Otherwise, the rest is straightforward and we showcase what we are looking for.

$\square$

1d

When n is very large, show that Var$(\bar{Y}) \approx \rho\sigma_Y^2$.

Solution Attempt: Take our expression from above,

$$Var(\bar{Y}) = \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right)\rho\sigma_Y^2$$

$$\frac{\sigma_Y^2}{n} \to 0 \quad \text{as} \quad n \to \infty$$

$$\frac{n-1}{n} \to 1 \quad \text{as} \quad n \to \infty$$

$Var(\bar{Y}) \to \rho\sigma_Y^2$ for large n and $Var(\bar{Y}) \approx \rho\sigma_Y^2$.

$\square$

Problem 2

On the companion website, you will find the spreadsheet `Age_HourlyEarnings`, which contains the joint distribution of age (`Age`) and average hourly earnings (`AHE`) for 25-to-34 year old full-time workers in 2012 with an education level that exceeds a high school diploma. Use this joint distribution to carry out the following exercises.

2a

Compute the marginal distribution of `Age`.

Solution Attempt: For this part and future parts, I've converted the dataset into a clean `.csv` file and saved it as `df`.

```
df.sum(axis=0)
```

```
25    0.084967
26    0.090789
27    0.085471
28    0.093518
29    0.103331
30    0.104217
31    0.103166
32    0.108084
33    0.108386
34    0.114542
dtype: float64
```

This above shows us the values of the `Age` marginal distribution. Notice that this sums to one.

```
age_marginal = df.sum(axis=0)
age_marginal.sum()
```

```
0.9964715700000001
```

□

2b

Compute the mean of `AHE` for each value of `Age`; that is, compute $E(\texttt{AHE} \mid \texttt{Age} = 25)$ and so forth.

Solution Attempt: For each age $a \in \{25, \ldots, 34\}$

$$E(\texttt{AHE} \mid \texttt{Age} = a) = \sum_y y \cdot P(\texttt{AHE} = y \mid \texttt{Age} = a) = \frac{\sum_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)}{\sum_y P(\texttt{AHE} = y, \texttt{Age} = a)}$$

```
means = {}
for age in df.columns:
  col = df[age]
  num = (col.index * col).sum()
  den = col.sum()
  means[age] = num / den

age_means = pd.DataFrame(
  list(means.items()),
  columns = ['Age', 'E(AHE | Age)']
)
print(age_means)
```

```
   Age  E(AHE | Age)
0   25     17.634440
1   26     19.019161
2   27     19.704933
3   28     20.182865
4   29     21.142915
5   30     21.811767
6   31     22.724216
7   32     23.691860
8   33     23.279501
9   34     23.984285
```

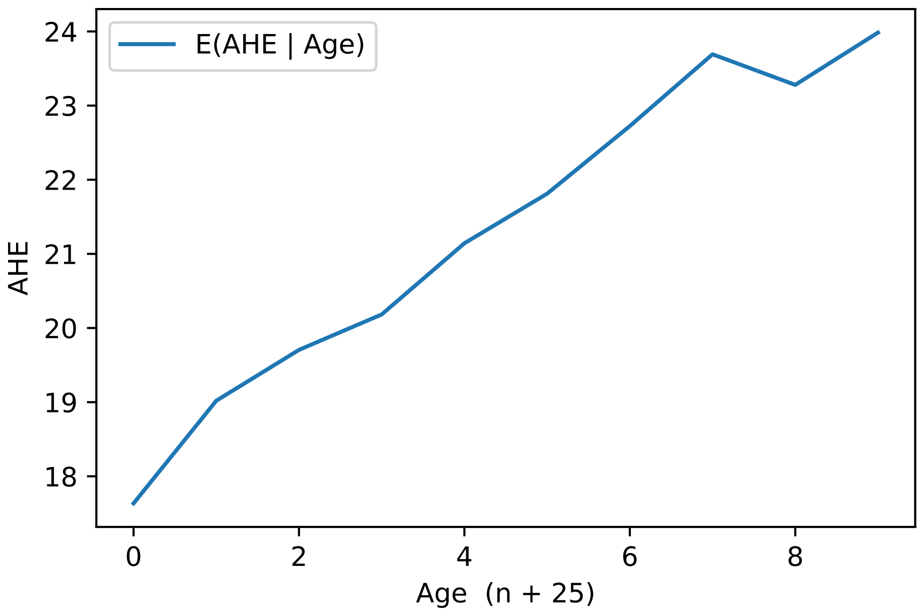
□

2c

Compute and plot the mean of `AHE` versus `Age`. Are average hourly earnings and age related? Explain.

Solution Attempt:

```
age_means.plot(
  xlabel = 'Age  (n + 25)',
  ylabel = 'AHE'
)
```



$\square$

2d

Use the law of iterated expectations to compute the mean of `AHE`; that is, compute $E(\texttt{AHE})$.

Solution Attempt: Here is what we are looking for,

$$E(\texttt{AHE}) = E(E(\texttt{AHE} \mid \texttt{Age})) = \sum_a \underbrace{E(\texttt{AHE} \mid \texttt{Age} = a)}_{m(a)} \underbrace{P(\texttt{Age} = a)}_{P_A}$$

where

$$m(a) = \frac{\sigma_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)}{\sum_y P(\texttt{AHE} = y, \texttt{Age} = a)} \quad \text{and} \quad P_a = \sum_y P(\texttt{AHE} = y, \texttt{Age} = a)$$

And, we can verify directly that

$$E(\texttt{AHE}) = \sum_a \sum_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)$$

```
# P(Age=a)
P_age = df.sum(axis=0)

# m(a) = E[AHE | Age=a]
means = {}
for a in df.columns:
    col = df[a]
    means[a] = (col.index * col).sum() / col.sum()
m = pd.Series(means)

# E[AHE] via law of iterated expectations
E_AHE = (m * P_age).sum()
print("E[AHE] via LIE =", E_AHE)
```

```
E[AHE] via LIE = 21.430079466
```

```
E_AHE_direct = (df.mul(df.index, axis=0)).sum().sum()
print("E[AHE] direct  =", E_AHE_direct)
```

```
E[AHE] direct  = 21.430079466
```

Great, we get the same things.

□

2e

Compute the variance of AHE.

Solution Attempt: The direct definition,

$$\mu = E(\texttt{AHE}), \quad E(\texttt{AHE}^2) = \sum_y y^2 \cdot P(\texttt{AHE} = y), \quad \boxed{Var(\texttt{AHE}) = E(\texttt{AHE}^2) - \mu^2}$$

Law of total variance says,

$$\boxed{Var(\texttt{AHE}) = E(Var(\texttt{AHE} \mid \texttt{Age})) = Var(E(\texttt{AHE} \mid \texttt{Age}))}$$

We can do this directly, or with the law of total variance. Let's try both see if we get the same.

```python
# P(AHE = y)
P_ahe = df.sum(axis=1)

# mean and second moment
mu = (P_ahe.index * P_ahe).sum()
E2 = ((P_ahe.index**2) * P_ahe).sum()

var_ahe = E2 - mu**2
print('Var(AHE) =', var_ahe)
```

```
Var(AHE) = 165.45054164692527
```

And now with law of total variance,

```python
# weights over Age
P_age = df.sum(axis=0)

m, v = {}, {}
for a in df.columns:
  col = df[a]
  p = col / col.sum()              # P(AHE | Age=a)
  ma = (p.index * p).sum()         # E(AHE | Age=a)
  E2a = ((p.index**2) * p).sum()   # E(AHE^2 | Age=a)
  m[a] = ma
  v[a] = E2a - ma**2               # Var(AHE | Age=a)

m = pd.Series(m)
v = pd.Series(v)

mu = (m * P_age).sum()             # E(AHE)
var_ahe_LTV = (v * P_age).sum() + ((m - mu)**2 * P_age).sum()
print('Var(AHE) via LTV =', var_ahe_LTV)
```

```
Var(AHE) via LTV = 163.83011614687115
```

There seems to be some error in rounding, truncating, or data types but we get essentially the same thing. For grading purposes, go with the first one.

□

2f

Compute the covariance between AHE and Age.

Solution Attempt: Let $Y = \texttt{AHE}$ and $A = \texttt{Age}$.

$$Cov(Y, A) = E(YA) - E(Y)E(A)$$

From the joint table $P(Y = y, A = a)$,

$$E(Y) = \sum_y yP_Y(y), \quad E(A) = \sum_a aP_A(a), \quad E(YA) = \sum_y \sum_a yaP(y, a)$$

```
# I'm going to try to fix the previous data type issue.
# Ensure numeric Age labels
df.columns = df.columns.astype(int)

# marginals
P_ahe = df.sum(1)     # P(Y)
P_age = df.sum(0)     # P(A)

# means
E_Y = (P_ahe.index * P_ahe).sum()
E_A = (P_age.index * P_age).sum()

# cross-moment E(YA) from joint
E_YA = df.mul(df.columns, 1).mul(df.index, 0).to_numpy().sum()

# covariance
cov_YA = E_YA - E_Y * E_A
print('Cov(AHE, Age) =', cov_YA)
```

```
Cov(AHE, Age) = 7.855846033653734
```

□

2g

Compute the correlation between AHE and Age.

Solution Attempt:

```
Var_Y = ((P_ahe.index ** 2) * P_ahe).sum() - E_Y ** 2
Var_A = ((P_age.index ** 2) * P_age).sum() - E_A ** 2
corr = cov_YA / (Var_Y ** 0.5 * Var_A ** 0.5)
print('Corr(AHE, Age) =', corr)
```

```
Corr(AHE, Age) = 0.18226305899108558
```

□

2h

Relate your answers in parts (f) and (g) to the plot you constructed in (c).

Solution Attempt: The plot in (c) shows $E(\texttt{AHE} \mid \texttt{Age})$ rising roughly from $18 at 25 to roughly $24 at 34, so `AHE` and `Age` (and a positive correlation, albeit not huge).

Your variance result and the law of total variance say

$$Var(\texttt{AHE}) = E(Var(\texttt{AHE} \mid \texttt{Age})) + Var(E(\texttt{AHE} \mid \texttt{Age})).$$

The plot reflects the second term (the between-age spread of the conditional means), which is small relative to the total variance— so most dispersion in `AHE` within age groups, not across ages.

So, basically, the plot explains the direction (positive association), while the variance/cov explaisn the magnitude.

□.

Problem 3

Let $X_1, \ldots, X_n$ be a random sample from the Bernoulli distribution with parameter $p = 0.78$ and $n = 25$. Generate $M = 100$ such random draws (there will be $M \times n = 2500$ random variables generated). For each random draw calculate a sample mean $\bar{X}^{(j)}$, where $j = 1, \ldots, M$.

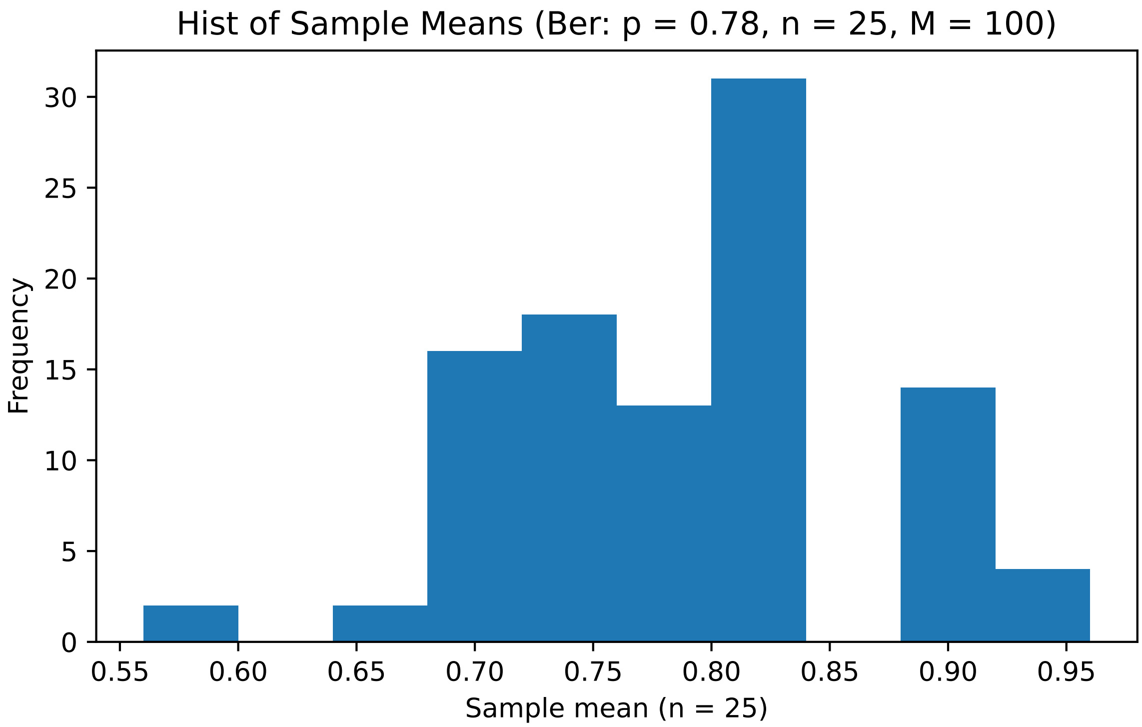
3a

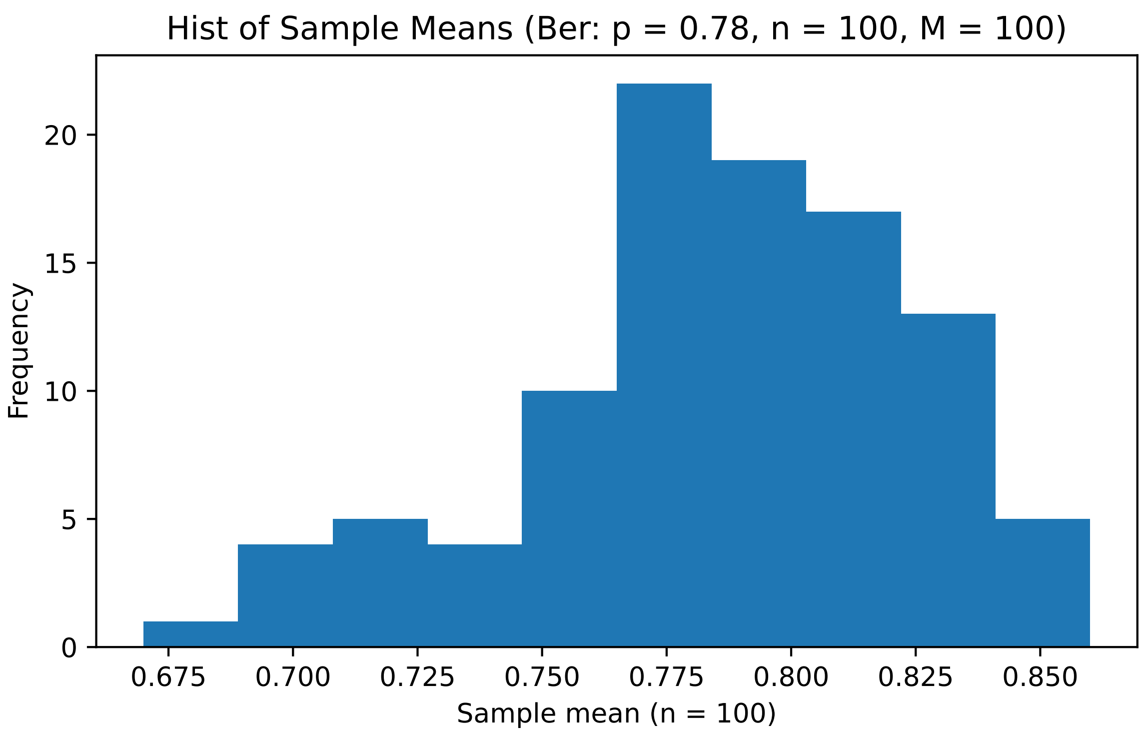
Draw a histogram $\bar{X}^{(1)}, \ldots \bar{X}^{(M)}$. Repeat the same for $n = 100$. Discuss the Law of Large Numbers.

Solution Attempt:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# parameters
p = 0.78
M = 100
rng = np.random.default_rng(42)

def simulate_means(n, M, p, rng):
  draws = rng.binomial(1, p, size = (M, n))
  return draws.mean(1)

mean_25 = simulate_means(25, M, p, rng)
mean_100 = simulate_means(100, M, p, rng)

plt.figure(figsize = (7,4))
plt.hist(mean_25, bins = 10, )
plt.xlabel('Sample mean (n = 25)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 25, M = 100)')

plt.figure(figsize = (7,4))
plt.hist(mean_100, bins = 10)
plt.xlabel('Sample mean (n = 100)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')
```

```
Text(0.5, 1.0, 'Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')
```


Hist of Sample Means (Ber: p = 0.78, n = 25, M = 100)
Frequency
Sample mean (n = 25)
0
5
10
15
20
25
30
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95


Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)
Frequency
Sample mean (n = 100)
0
5
10
15
20
0.675
0.700
0.725
0.750
0.775
0.800
0.825
0.850

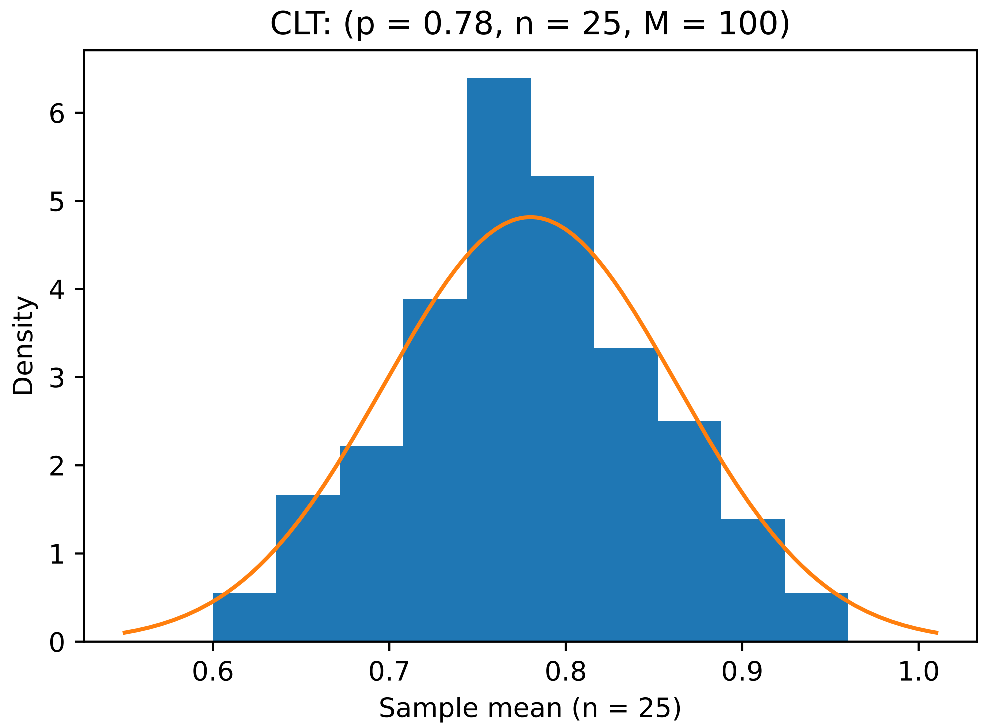
□

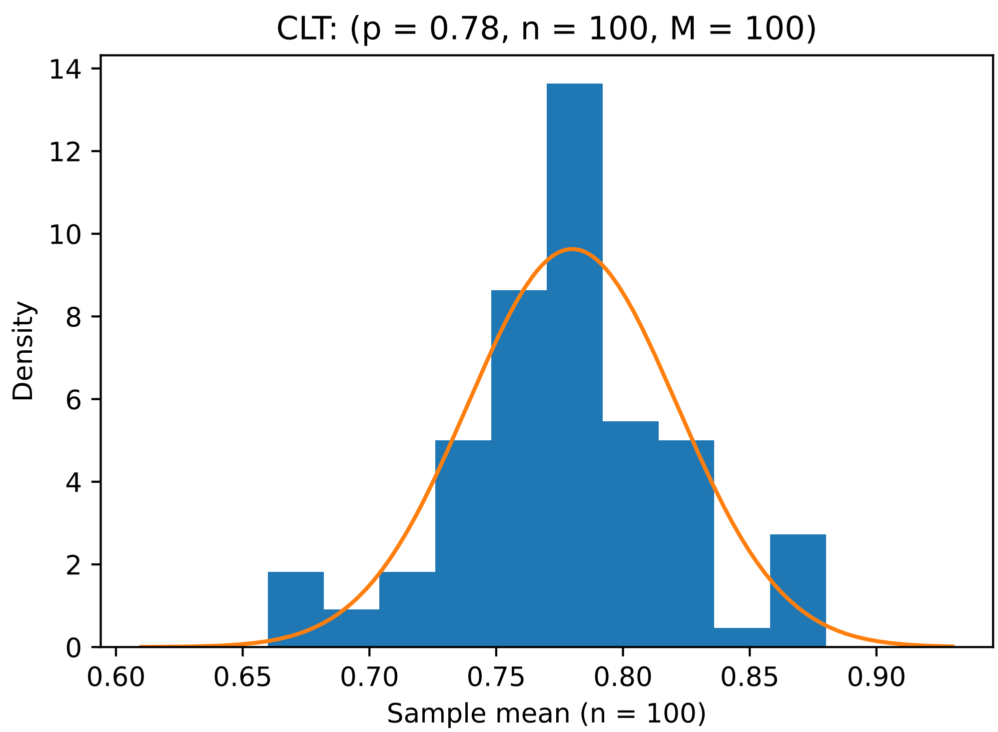
3b

For $n = 25$ and $n = 100$ draw corresponding histograms to illustrate the Central Limit Theorem. Dicuss.

Solution Attempt:

```
from math import sqrt, pi, exp

p = 0.78
M = 100
rng = np.random.default_rng(7)

def simulate_means(n):
    x = rng.binomial(1, p, size=(M, n)).mean(1)
    return x

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

for n in (25, 100):
    means = simulate_means(n)
    mu = p
    sd = sqrt(p * (1 - p) / n)

    plt.figure(figsize = (6, 4))
    # histogram of sample means (density so total area ~1)
    plt.hist(means, bins = 10, density = True)
    # theoretical normal curve
    xs = np.linspace(min(means) - 0.05, max(means) + 0.05, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"CLT: (p = {p}, n = {n}, M = {M})")
```


CLT: (p = 0.78, n = 25, M = 100)
Density
0
1
2
3
4
5
6
0.6
0.7
0.8
0.9
1.0
Sample mean (n = 25)


CLT: (p = 0.78, n = 100, M = 100)
Density
0
2
4
6
8
10
12
14
0.60
0.65
0.70
0.75
0.80
0.85
0.90
Sample mean (n = 100)

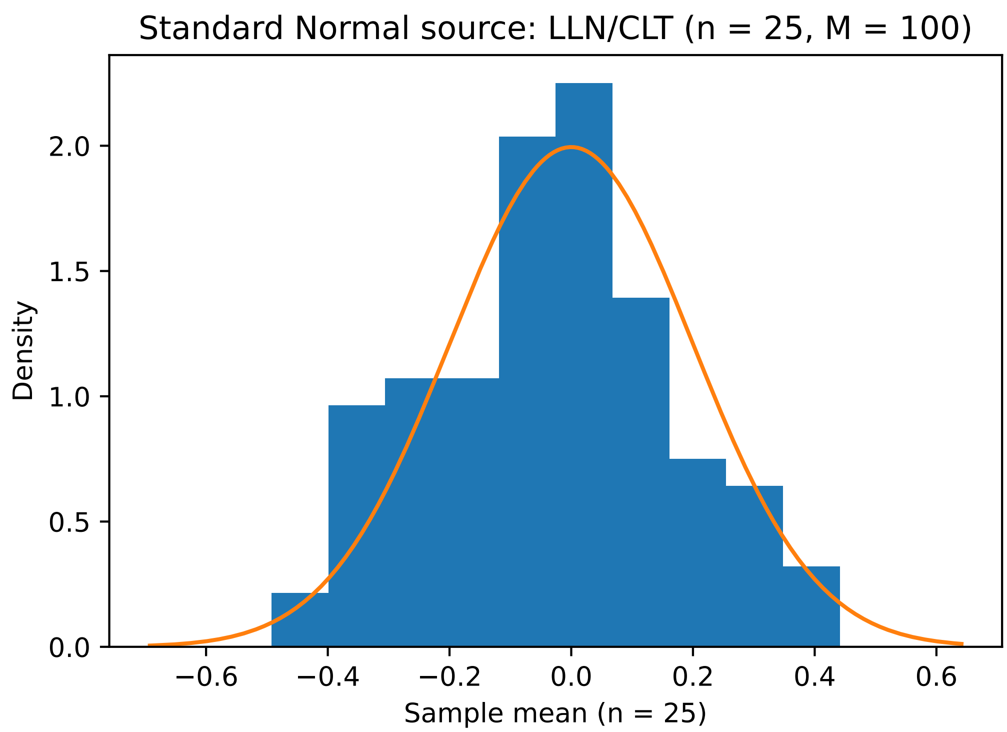
□

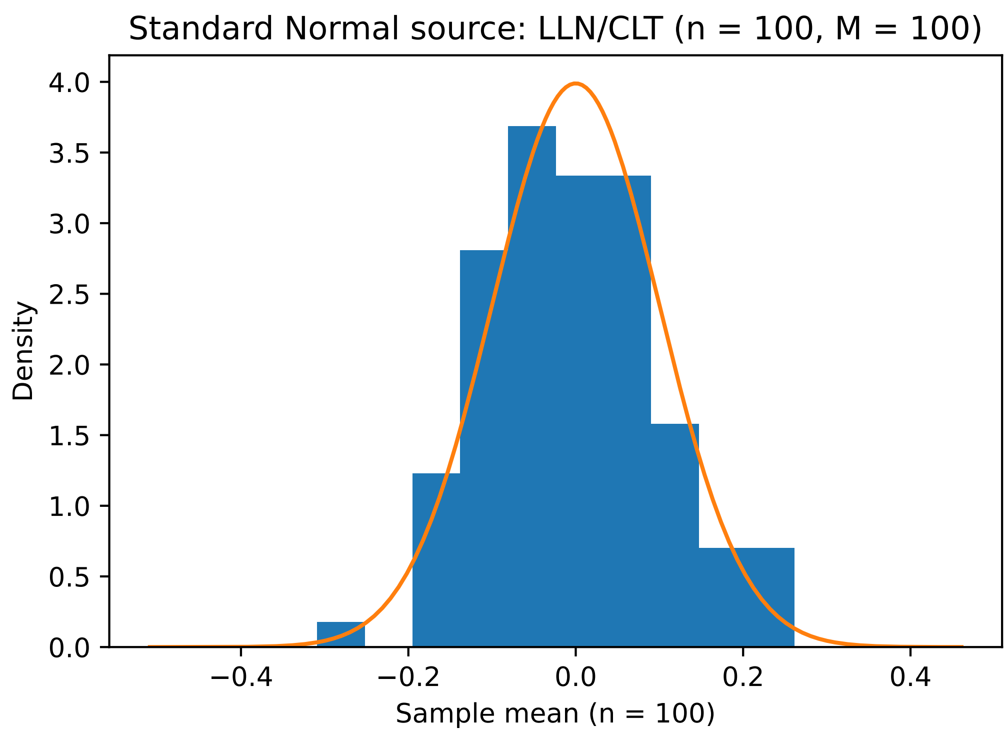
3c

Repeat (a) and (b) for the random draw from the standard normal distribution.

Solution Attempt:

```
M = 100
rng = np.random.default_rng(7)

def simulate_means_normal(n):
    # M x n draws from N(0,1); take sample means
    return rng.standard_normal(size=(M, n)).mean(axis=1)

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

for n in (25, 100):
    means = simulate_means_normal(n)
    mu, sd = 0.0, 1.0 / sqrt(n)           # exact: mean 0, Var = 1/n

    plt.figure(figsize = (6, 4))
    plt.hist(means, bins = 10, density = True)
    xs = np.linspace(means.min() - 0.2, means.max() + 0.2, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"Standard Normal source: LLN/CLT (n = {n}, M = {M})")
    plt.show()
```


Standard Normal source: LLN/CLT (n = 25, M = 100)
Density
Sample mean (n = 25)
2.0
1.5
1.0
0.5
0.0
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


Standard Normal source: LLN/CLT (n = 100, M = 100)
Density
Sample mean (n = 100)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.4
−0.2
0.0
0.2
0.4

$\square$

3d

Compare your answers for the Bernoulli and the standard normal distribution for the LLN and the CLT.

Solution Attempt:

LLN (convergence of $\bar{X}$ to the true mean)

- $Ber(p = 0.78)$: $\bar{X} \to 0.78$. Spread is approx. $\sqrt{p(1-p)/n} = \sqrt{0.1716/n}$.
- Stadard Normal: $\bar{X} \to 0$. Spread is approx. $\sqrt{1/n}$.

Both concentrate at their means and shrink like $1/\sqrt{n}$. Normal shrinks from a larger variance (1 vs. 0.1716), so its histogram are wider for the same n.

CLT (shape of the sampling distribution):

- $Ber(p = 0.78)$: Approx. normal for moderate n; with $p = 0.78$ the $n = 25$ histogram can show mild skew effects (bounded in $[0, 1]$), but by $n = 100$ it's close to normal.
- Standard Normal: $\bar{X}$ is exactly $\mathcal{N}(0, 1/n)$ for any n so the curve matches quickly.